COMSATS University Islamabad, *Lahore Campus*

Department of Computer Engineering

Quiz 2 – SPRING 2026

Course Title:	Digital System Design		Course Code:	CPE244	Credit Hours: 4(3,1)
Course Instructor:	Dr. Engr. M Babar Ali		Program Name:	BSc in Computer Engineering	
Semester:	7th	Batch:	FA23	Section:	A/B
Date:		**Time Allowed:**	**20 minutes**	**Maximum Marks:**	**10**
Student's Name:		Reg. No.	CUI/ - - /LHR		

Important Instructions / Guidelines:

- Write your name and registration number clearly in respective fields.
- Mobile phones are not allowed in the classroom.
- Do not give multiple answers to a question.

Question: [CLO1, PLO3 – C5] **(Marks 10)**

Design a synchronous digital system consisting of 8-bit data input I, three load control inputs L_A, L_B, and L_C, a rotation control input R, and a clock input CLK. The system has three 8-bit outputs, A, B, and C, representing the current contents stored in the system.

The operation of the system is as follows: When $L_A = 1$, system stores input data I and outputs it on output A. Similarly, when $L_B = 1$ or $L_C = 1$, system stores the input data $I$and displays it on output B or C, respectively. If any of the load control input L_A, L_B or L_C is 0, the system retains its all previously stored values. When $R = 1$, the stored contents rotate simultaneously such that output A displays the previous value on output B, the output B displays the previous value on C, and the output C displays the previous value on A. The rotation control input R has the highest priority over the load control inputs. Therefore, when $R = 1$, rotation occurs regardless of the values of L_A, L_B, and L_C. When $R = 0$, the registers operate according to their respective load control inputs.

Use four step Register Design Process for this design where all register operations occur synchronously on the active edge of the clock signal CLK.